Bibliography

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V^{th} **Semester**
Report

FPGA Based PPG Waveform Analysis and Disease Prediction

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the requirements of the In Semester Assessment for course PR1: Mini Proejct 1

Submitted by

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CERTIFICATE

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Abstract

Cardiovascular diseases (CVDs) are the leading cause of global mortality, contributing to nearly 32% of deaths worldwide. This work presents a real-time Photoplethysmography (PPG) acquisition and cardiovascular risk detection system implemented on a Xilinx Artix-7 FPGA, targeting portable and low-latency health monitoring applications.

A custom TCRT1000 infrared PPG sensor with optimized analog signal conditioning ensures high-fidelity waveform acquisition through cascaded high-pass and low-pass filtering with an overall gain of 301. The sampled signal is processed entirely on hardware using parallelized modules for adaptive peak detection, statistical feature extraction (mean, RMS, standard deviation), and computation of clinically relevant cardiovascular indicators such as systolic/diastolic amplitude, peak-to-peak variation, and BPM.

A 30-tree Random Forest classifier, developed using HLS4ML and Conifer, was quantized to 12-bit fixed-point representation and integrated as a custom IP core. The hardware accelerator achieves 96.32% accuracy in myocardial infarction detection with only 26 clock cycles of inference latency — corresponding to 260 ns at 100 MHz, which is over four orders of magnitude faster than software execution.

The complete system utilizes 34.16% of LUTs and 24.44% of flip-flops, confirming efficient FPGA resource usage. Real-time outputs are displayed via dual LCD modules and validated externally through UART waveform streaming. Experimental evaluation demonstrates accurate heart-rate computation and robust disease classification, establishing FPGA-based acceleration as a promising approach for next-generation point-of-care cardiovascular diagnostics.

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Chapter 1

Introduction

Overview

Cardiovascular diseases (CVDs) are the leading cause of mortality worldwide, responsible for approximately 32% of all global deaths. In recent years, the healthcare system has been steadily shifting from reactive treatment to preventive care. Continuous monitoring of vital physiological parameters plays a critical role in the early identification of cardiac abnormalities, allowing timely intervention, improving patient outcomes, and reducing the burden on the medical infrastructure.

Various clinical and diagnostic techniques exist to assess cardiac health; however, many of these approaches are invasive and can involve surgical procedures that pose discomfort and risk to patients. Consequently, non-invasive diagnostic techniques have gained substantial importance, providing reliable physiological data without the need for penetration or intrusive instrumentation. Among these, Photoplethysmography (PPG) has emerged as a widely adopted method that utilizes light-based sensing to monitor blood volume changes and extract heart-related information simply by placing a sensor on the skin.

Although PPG has traditionally been used for basic heart-rate measurement, its evolution to predictive diagnostics demands high-accuracy signal interpretation. Extracting clinically meaningful features from PPG signals requires complex algorithms for noise reduction, waveform analysis, feature extraction, and disease classification. These computational requirements raise the need for high processing speed, low latency, and improved power efficiency. This project addresses these challenges by implementing a PPG waveform analysis and disease prediction system on a Field Programmable Gate Array (FPGA), leveraging hardware parallelism to enable high-speed, real-time diagnosis.

Problem Statement: The Software Bottleneck

Most current portable health devices are based on microcontrollers (MCUs) or general-purpose processors. These architectures operate on a sequential execution model, fetching and decoding instructions one instruction at a time. Although sufficient for basic data logging, this approach creates a bottleneck when applied to complex, real-time disease prediction.

The primary limitations of software-based approaches include the following:

- **Latency and Jitter:** Sequential processing introduces unpredictable delays, par-

ticularly when multiple algorithms (filtering, peak detection, and classification) run concurrently. In critical health monitoring, deterministic timing is essential.

- **Computational Overhead:** High-precision digital filtering (such as FIR filters) requires thousands of multiplication and accumulation operations per second. General-purpose ALUs are not optimized for such continuous mathematical data streams, resulting in reduced efficiency and slower response times.
- **Noise Sensitivity:** Real-world PPG signals are heavily corrupted by motion artifacts. Software filters struggle to clean signals dynamically without introducing phase lag, which can distort the very morphological features required for an accurate diagnosis.

1.3 Proposed Solution: FPGA-Based Hardware Acceleration

To overcome the limitations of software-based PPG processing, this project implements disease prediction using Field Programmable Gate Array (FPGA) hardware acceleration. Unlike sequential processors, FPGAs leverage massive parallelism, enabling multiple signal-processing tasks to be executed simultaneously.

In the proposed architecture, noise filtration, feature extraction, and decision logic are mapped onto dedicated hardware blocks. Each sub-system operates concurrently rather than sequentially. For example, the FPGA's DSP slices efficiently handle filtering, while finite state machines (FSMs) perform real-time peak detection without affecting signal acquisition.

This approach ensures deterministic latency, meaning that the time between signal sampling and disease prediction remains constant and minimal. By designing custom hardware specifically tailored for PPG morphology analysis, this project aims to demonstrate a reliable, high-speed alternative to traditional software-based medical monitoring systems, therby enabling the next generation of point-of-care diagnostic devices.

Chapter 2

Literature Review

Photoplethysmography (PPG) has emerged as a powerful non-invasive technique for cardiovascular health monitoring and disease detection. This section reviews recent advances in PPG signal processing and implementation platforms, examining six significant contributions spanning hardware sensor design, signal processing algorithms, and disease prediction systems.

2.1 Signal Processing and Feature Extraction

Effective PPG-based disease detection requires robust signal processing methodologies. Vadrevu and Manikandan [2] developed a comprehensive pulse onset and peak detection method utilizing stationary wavelet transform (SWT), multiscale sum and product (MSP), and Shannon entropy envelope extraction. Their approach achieved 99.66% sensitivity, 99.90% positive predictivity, and 99.55% overall accuracy across 116,255 beats from multiple databases. The method maintained robustness even at signal-to-noise ratios as low as 10 dB, eliminating the need for search-back algorithms while enabling accurate pulse rate variability analysis.

Kong and Hedayatipour [5] developed a comprehensive PPG signal processing pipeline utilizing Dynamic Time Warping (DTW)-based signal quality indexing. From 1.83 million initial segments, their quality assessment retained 83,597 high-quality 60-second segments from the MIMIC-III database. Using the pyPPG toolbox, they extracted 102 comprehensive pulse wave features including time-domain, amplitude-based, and derivative-based characteristics, providing a valuable resource for cardiovascular health assessment and AI model development.

Ihsan et al. [4] examined three feature extraction approaches for coronary heart disease (CHD) detection: RR interval features, heart rate variability (HRV) features, and time-domain features. Analysis of 28 CHD patients and 30 healthy subjects revealed that HRV features achieved superior performance with 94.44% accuracy, 100% sensitivity, and 90.9% specificity using decision tree classification. Shannon entropy emerged as the most critical discriminative feature, demonstrating that feature quality significantly outweighs feature quantity.

2.2 Hardware Design and Sensor Development

Peng et al. [3] addressed sensor design challenges through Monte Carlo simulation of light propagation in multi-layered tissue. Their reflective-mode sensor design, featuring a central photodiode surrounded by four LEDs, achieved optimal performance with source-detector separation of 0.3-0.4 cm. This configuration successfully captured high-quality PPG signals from five diverse anatomical locations: chest, forehead, palm, wrist, and back of wrist.

Wang et al. [6] developed a wearable finger-based PPG device for non-invasive blood glucose monitoring. Their system employed dual-wavelength measurement (660 nm and 880 nm) combined with a multi-layer perceptron neural network. Testing on 4,042 PPG-glucose data pairs yielded clinically acceptable results: 96% of predictions in Clarke Error Grid Region A and 99.33% in combined Regions A and B, with low-cost hardware (<10 USD) and extended battery life (25+ hours).

2.3 FPGA-Based Implementation: A Unique Approach

While previous studies employed software-based implementations using MATLAB [2], Python [4, 5], or microcontroller platforms [6], Chowdhury et al. [1] pioneered the first FPGA-based multiclass cardiovascular disease detection system utilizing PPG signals. Their implementation on Xilinx Zynq UltraScale platform integrates hardware-accelerated FIR filters for preprocessing (high-pass at 0.5 Hz, low-pass at 15 Hz), followed by parallel feature extraction and seven simultaneous SVM classifiers for multiclass disease detection.

The system classifies PPG signals into seven categories: normal, diabetes mellitus type 2, cerebral infarction, cerebrovascular disease, stage 1-2 hypertension, hypertension with DM2, and prehypertension with DM2. Testing on 331 high-quality recordings from 219 subjects achieved 79.83% overall accuracy with 1.403 W total power consumption.

The FPGA implementation offers distinct advantages over conventional approaches. Hardware parallelism enables simultaneous execution of filtering, feature extraction, and classification operations, eliminating sequential processing bottlenecks. Dedicated hardware pipelines provide deterministic latency crucial for real-time health monitoring, while reconfigurability supports algorithm evolution and multi-modal physiological monitoring. This distinguishes FPGA implementation from software systems, which introduce significant processing latency and non-deterministic timing, and from SoC platforms, which lack reconfigurability and parallel processing capabilities.

2.4 Comparative Analysis

Table 2.1 provides a comprehensive comparison of the reviewed literature across implementation platform, dataset characteristics, signal processing methodologies, feature extraction approaches, classification techniques, and performance metrics.

Analysis reveals a significant research gap in hardware-accelerated PPG processing systems. Software implementations dominate current research, offering development flexibility but presenting fundamental limitations for real-time clinical applications. Vadrevu and Manikandan's [2] algorithm required 426.6 ms to process an 8.192-second segment in MATLAB, while Kong and Hedayatipour [5] processed 83,597 segments in 17 hours. Microcontroller implementations [6] achieve portability and low cost but typically rely on offline

Table 2.1: Comparative Analysis of PPG-Based Disease Detection Systems

Study	**Platform**	**Dataset Size**	**Signal Processing**	**Feature Set**	**Classifier**	**Accuracy**
Chowdhury et al. [1]	**FPGA (Zynq UltraScale)**	331 recordings, 219 subjects	FIR filters (HPF 0.5Hz, LPF 15Hz)	9 statistical + 2 physical	Multiclass SVM	79.83%
Vadrevu & Manikandan [2]	Software (MATLAB)	116,255 beats	SWT, MSP, Shannon entropy, GDF	Peak/onset timing	Detection algorithm	99.66% Se
Peng et al. [3]	Hardware sensor	5 body sites	FIR Butterworth, Monte Carlo	Source-detector separation	N/A	High quality
Ihsan et al. [4]	Software (Python)	28 CHD, 30 healthy	FIR filter, HRV analysis	6 HRV features	Decision Tree	94.44%
Kong & Hedayatipour [5]	Software (Python)	83,597 segments	DTW-based SQI, pyPPG	102 pulse features	N/A	N/A
Wang et al. [6]	Wearable (STM32)	4,042 PPG-glucose pairs	Butterworth LPF (5Hz)	Dual-wavelength intensity	MLP	99.33% CEG A+B

processing for complex analyses.

The FPGA-based approach [1] addresses these limitations through hardware-software co-design, enabling real-time multiclass disease detection with deterministic performance. This represents a paradigm shift from software-centric to hardware-accelerated PPG analysis, offering a pathway toward next-generation wearable health monitoring systems combining reconfigurable hardware flexibility with dedicated processing pipeline performance.

2.5 Summary

The reviewed literature demonstrates PPG technology maturity for cardiovascular health monitoring, achieving accuracies from 79.83% to 99.66% across diverse applications. Signal quality assessment using DTW-based filtering [5] and adaptive thresholding [2] proves effective for noise rejection. HRV features, particularly Shannon entropy [4], provide superior discriminative power for cardiovascular disease classification. Optimal source-detector geometry [3] ensures reliable signal acquisition across multiple anatomical sites.

The transition from software platforms [2, 4–6] to hardware implementations represents a crucial step toward clinical deployment. The FPGA-based implementation [1] offers a unique combination of real-time processing capability, reconfigurability, and power efficiency, addressing fundamental software implementation limitations and positioning FPGA technology as an enabling platform for autonomous cardiovascular health monitoring systems.

Chapter 3

Project Objectives

The primary objectives of this project are as follows:

- To develop an accurate and noise-free Photoplethysmography (PPG) sensor.
- To correctly sample and denoise the PPG waveform to obtain a clean signal suitable for analysis.
- To apply signal-processing techniques to extract cardiovascular health features from the PPG waveform, including:
  - Heart Rate
  - Interbeat Interval
  - Average Systolic Amplitude
  - Average Diastolic Amplitude
  - Peak-to-Peak
  - Root Mean Square (RMS)
  - Standard Deviation
  - Mean
- To implement a neural network model capable of predicting heart disease from extracted PPG features.
- To display and present the final results to the user.

Chapter 4

Theory

4.1 The Science Behind Photoplethysmography

Photoplethysmography (PPG) is an optical technique used to detect volumetric changes in blood in peripheral circulation. It operates on the principle of light absorption, wherein a light-emitting diode (LED) illuminates tissue (such as the fingertip, wrist, or earlobe), and a photodetector measures the variations in light intensity corresponding to changes in blood perfusion volume.

The PPG waveform consists of two primary components:

- **AC Component:** The pulsatile component produced due to the cardiac cycle (systole and diastole).
- **DC Component:** A slowly varying baseline affected by respiration, thermoregulation, and sympathetic nervous system activity.

The AC component provides the morphology of the blood flow and allows extraction of clinically useful features such as the systolic peak, diastolic peak, dicrotic notch, and pulse width. These parameters are strongly correlated with arterial stiffness, vascular aging, and blood pressure. Thus, PPG morphology is a valuable non-invasive method for early prediction of cardiovascular conditions such as hypertension and arrhythmia.

4.2 Signal Processing

Oversampling refers to sampling an analog signal at a frequency significantly higher than the Nyquist rate. This distributes quantization noise across a wider bandwidth. When combined with decimation (digital filtering followed by downsampling), high-frequency noise is attenuated while preserving the signal of interest.

This procedure reduces the noise floor, increases the Signal-to-Noise Ratio (SNR), and improves the Effective Number of Bits (ENOB). As a result, the system can capture subtle waveform features that would otherwise be lost in noise.

4.3 Algorithms Behind Feature Extraction

The following statistical and morphological features are extracted from the PPG waveform.

Basic Statistical Features

$$\text{mean} = \frac{1}{N}\sum_{i=1}^{N} x_i \tag{4.1}$$

$$\text{std} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \text{mean})^2} \tag{4.2}$$

Peak-Related Features

$$\text{peak_to_peak} = \text{Systolic} - \text{Diastolic} \tag{4.3}$$

$$\text{rms} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2} \tag{4.4}$$

Amplitude Features

$$\text{avg_systolic_amp} = \frac{1}{M}\sum_{j=1}^{M} S_j \tag{4.5}$$

where S_j represents the jth systolic peak and M is the total number of detected systolic peaks.

$$\text{avg_diastolic_amp} = \frac{1}{M}\sum_{j=1}^{M} D_j \tag{4.6}$$

where D_j represents the jth diastolic trough and M is the total number of detected diastolic troughs.

4.4 Numerical Techniques to Reduce Computational Overhead

Several numerical techniques are employed to reduce resource utilization during hardware implementation.

1. Square Root Block

The square root block implements numerical square root extraction using an iterative approximation technique. The non-restoring square root algorithm can be described as:

$$
\begin{aligned}
\text{remainder} &= x \\
\text{root} &= 0 \\
\text{for each bit position } i \text{ from MSB to LSB:} & \\
&\quad \text{trial} = \text{root} + 2^i \\
&\quad \text{if trial}^2 \leq \text{remainder:} \\
&\qquad \text{root} = \text{trial}
\end{aligned}
$$

2. Mean Block (Power-of-Two Division)

To optimize hardware efficiency, division is performed via right-shift operations, which is exact when the number of samples is a power of two.

$$\text{mean} = \frac{\sum_{i=1}^{N} x_i}{N} \tag{4.7}$$

If $N = 2^k$, then:

$$\text{mean} = (\sum_{i=1}^{N} x_i) \gg k \tag{4.8}$$

where $\gg k$ denotes a right-shift by k bits and $k = \log_2(N)$.

4.5 Machine Learning and its Quantization

The Random Forest classifier was optimized for FPGA deployment through a strategic quantization approach that balances computational accuracy with hardware resource efficiency. The model architecture, consisting of 30 decision trees with a maximum depth of 7, was originally trained using 64-bit floating-point precision as provided by the scikit-learn library. To enable efficient FPGA implementation on the Xilinx Artix-7 XC7A100T device, the model underwent conversion to fixed-point arithmetic using the Conifer framework, which specializes in translating ensemble machine learning models into High-Level Synthesis (HLS) compatible code.

The quantization strategy employed a 12-bit fixed-point representation specified as ap_fixed 12,5, allocating 5 bits for the integer portion and 7 bits for the fractional component. This configuration provides a numerical range of ± 16 with a precision granularity of approximately 0.0078 (equivalent to 1/128). The selection of 12-bit precision was made after careful analysis of the feature value distributions and model sensitivity, representing an optimal compromise between 10-bit representations (which showed insufficient precision for certain PPG signal features) and 16-bit representations (which would consume approximately 20-25% more FPGA lookup tables). The threshold comparisons within each decision tree node were maintained at the same `ap_fixed<12,5>` precision, while the tree score accumulation utilized an expanded `ap_fixed<18,8>` format to prevent overflow during the aggregation of predictions from multiple trees.

Resource optimization was further enhanced through architectural decisions in the HLS implementation. The configuration utilized serial I/O operations (`io_serial`) rather than

parallel processing, which dramatically reduces the instantiation of parallel comparator circuits and can decrease LUT utilization by 60-70% compared to fully unrolled implementations. A reuse factor of 16 was applied to the computation pipeline, enabling temporal multiplexing of hardware resources across multiple tree evaluations. Loop unrolling was deliberately disabled to prevent the duplication of decision node logic, and a pipeline interval of 2 clock cycles was introduced to balance throughput with resource consumption. These optimizations collectively reduced the estimated LUT requirement.

Chapter 5

System Design

5.1 System Overview

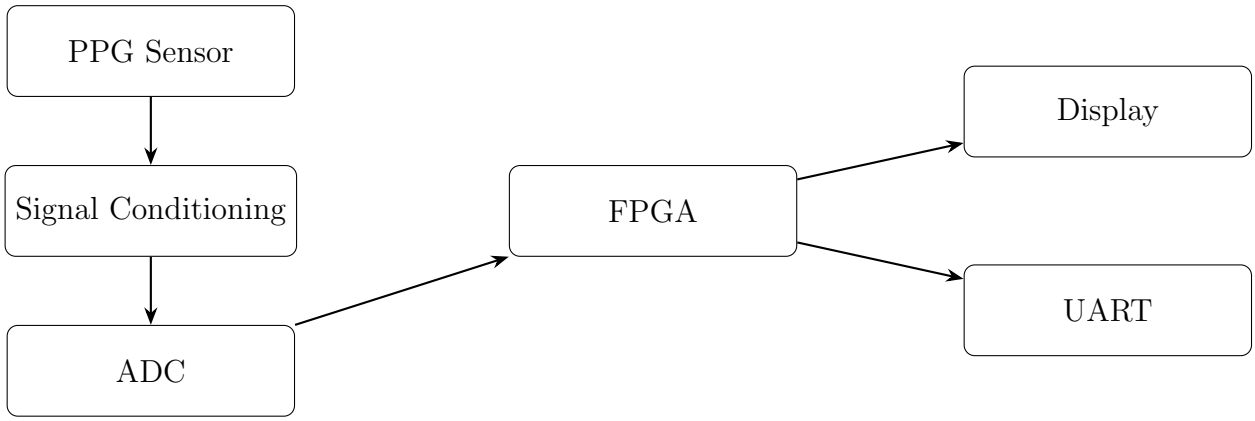

Figure 5.1: System block diagram .

Table 5.1: FPGA resource utilization summary.

Module / Function	**LUTs**	**Flip-Flops**	**DSP Slices**
Feature Extraction & Display Logic	9,791	18,537	5
Random Forest Algorithm	11,864	12,745	0
Total Requirement	**21,655**	**31,282**	**5**
Total Available	63,400	128,000	240
Utilization (%)	34.16	24.44	2.08

5.2 PPG Sensor and Signal Conditioning

PPG Sensor Operation

PPG sensors are optical devices that detect fluctuations in blood volume by analysing variations in reflected or transmitted light. Different wavelengths can be used to extract PPG

signals depending on tissue penetration and absorption characteristics. In this project, an infrared (IR) wavelength of **950 nm** is used, provided by the **TCRT1000 reflective optical sensor**.

The TCRT1000 houses both an IR LED and an IR photodetector that is tuned to detect only the emitted wavelength, ensuring minimal interference from ambient light. The sensor is powered by a **3.3 V supply** from the FPGA development board. To generate a waveform, the user must simply place a fingertip or wrist gently on the sensor surface, enabling optical coupling between tissue and detector.

Signal Conditioning Circuit

The raw output of the photodiode consists of a large **DC component** corresponding to baseline tissue absorption and a relatively small **AC component** corresponding to the pulsatile cardiac activity. Since the useful AC signal is often in the millivolt range, amplification and filtering are necessary to make the waveform suitable for sampling and analysis.

The signal conditioning stage performs the following operations:

- A **first-order passive high-pass filter** with a cutoff frequency of **0.7 Hz** removes the DC baseline while preserving the pulsatile AC component.
- A **first-order active low-pass filter** with a cutoff frequency of **2.34 Hz** eliminates high-frequency noise such as motion artefacts and ambient fluctuations.
- The amplifier stage provides a total gain of approximately **301**, ensuring that the AC waveform lies within the input range of the ADC.

The operational amplifier used for the signal conditioning stage is the **MCP6004**, a quad low-power CMOS op-amp. Only a single op-amp from the package is required for the PPG signal chain, ensuring low power consumption and compact circuitry.

5.3 Signal Processing and Feature Extraction

This section describes each functional module implemented on the FPGA for real-time PPG signal analysis, feature extraction, and system monitoring.

1) ADC Interface

The on-board 12-bit ADC samples the conditioned PPG waveform at 500Hz using an event-driven timer trigger. The ADC internally left-adjusts its 12-bit result into a 16-bit register. A right-shift and slice operation extracts the upper 12 valid bits. The ADC also provides an `EOC` (End of Conversion) signal that validates the sampled data before it is passed to the succeeding modules.

2) Mean Detector

The Mean Detector computes the moving average of the most recent 2048 samples (approximately 4 seconds of data). Each incoming 12-bit sample is added into a running accumulator

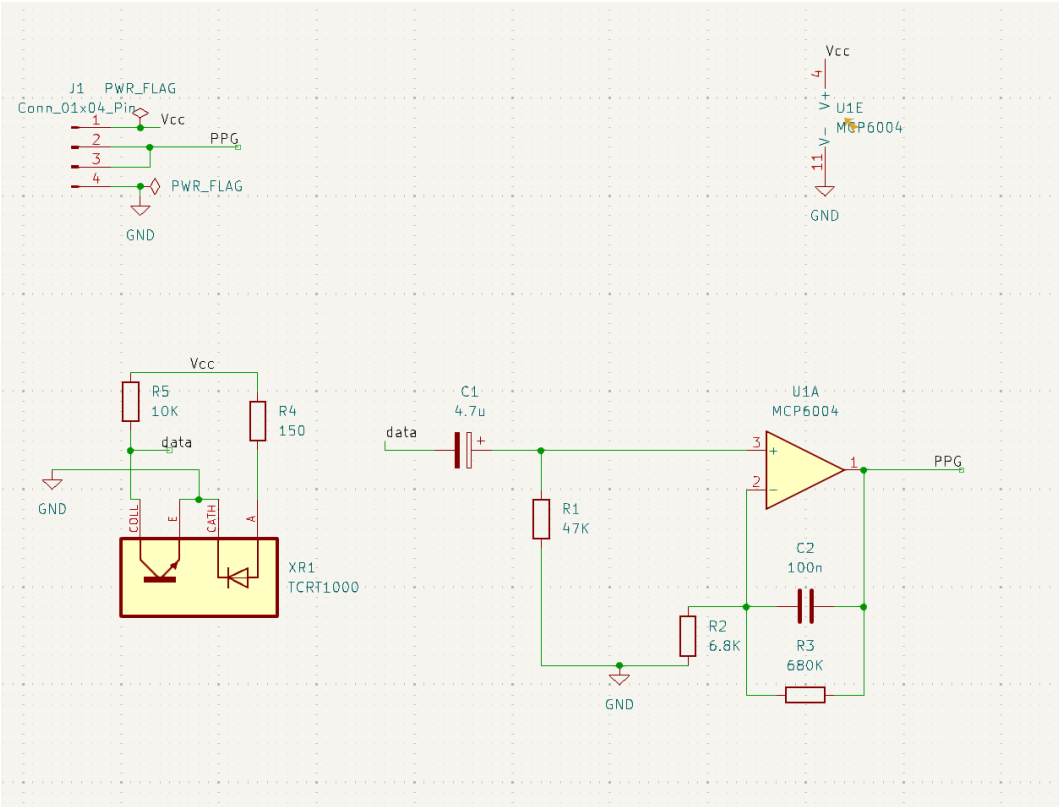

Figure 5.2: PPG sensor and Signal Conditioning Circuit.

while simultaneously discarding the oldest sample using a circular buffer. After 2048 samples, the accumulator value is right-shifted by 11 bits, achieving an efficient power-of-two division. The resulting mean value is forwarded to LCD 2 and the Machine Learning (ML) model.

3) RMS Detector

This module follows a process similar to the mean detector but first squares each 12-bit input sample to form a 24-bit value. The squared values are accumulated over 2048 samples and right-shifted by 11 bits. A hardware non-restoring square-root algorithm computes the final RMS value. This output is sent to LCD 2 and the ML model. A circular buffer ensures that only the latest 2048 samples contribute to the result.

4) Standard Deviation Calculator

The standard deviation module also maintains a 2048-sample circular buffer. It first computes the mean of these samples, then subtracts this mean from each stored value to obtain the deviation. Each deviation is squared, accumulated, and right-shifted by 11 bits. A non-restoring square-root unit then produces the final standard deviation. The output is used both by LCD 2 and the ML model.

5) Peak Detector

This module receives raw ADC samples and performs adaptive multiscale peak detection. By comparing each sample with its neighbouring samples and using adaptive thresholds, it identifies systolic and diastolic peaks. Peak amplitudes and validity signals are generated and forwarded to LCD 1 as well as the Peak-to-Peak Calculator.

6) Peak-to-Peak Calculator

This unit computes the difference between the systolic and diastolic amplitudes to obtain the peak-to-peak value. The result is displayed on LCD 1 and used in the ML model.

7) Timer Module

The timer counts clock cycles continuously and resets whenever the peak detector asserts a valid-peak signal. The timer output represents the time between heartbeats and is sent to the BPM Calculator.

8) BPM Calculator

The BPM Calculator divides a fixed scale factor by the timer value to compute the heart rate in beats per minute. Optimized hardware division ensures fast real-time operation. The calculated BPM is sent to LCD 1.

9) LCD Driver 1

This driver receives BPM, systolic amplitude, diastolic amplitude, and peak-to-peak values. It formats and prints them onto the 16×2 LCD with appropriate labels for real-time display.

10) LCD Driver 2

This driver receives the mean, RMS, standard deviation, and ML model output. It handles formatting and displays these processed statistical and diagnostic parameters on the second 16×2 LCD.

11) UART Driver

The UART Driver transmits raw PPG samples for external monitoring and debugging. Each 16-bit ADC sample is split into two 8-bit UART frames. A header byte `0xFF` precedes every transmitted pair to ensure safe synchronization during decoding. A FIFO buffer prevents sample loss and waveform mixing. The transmission baud rate is set to 9600.

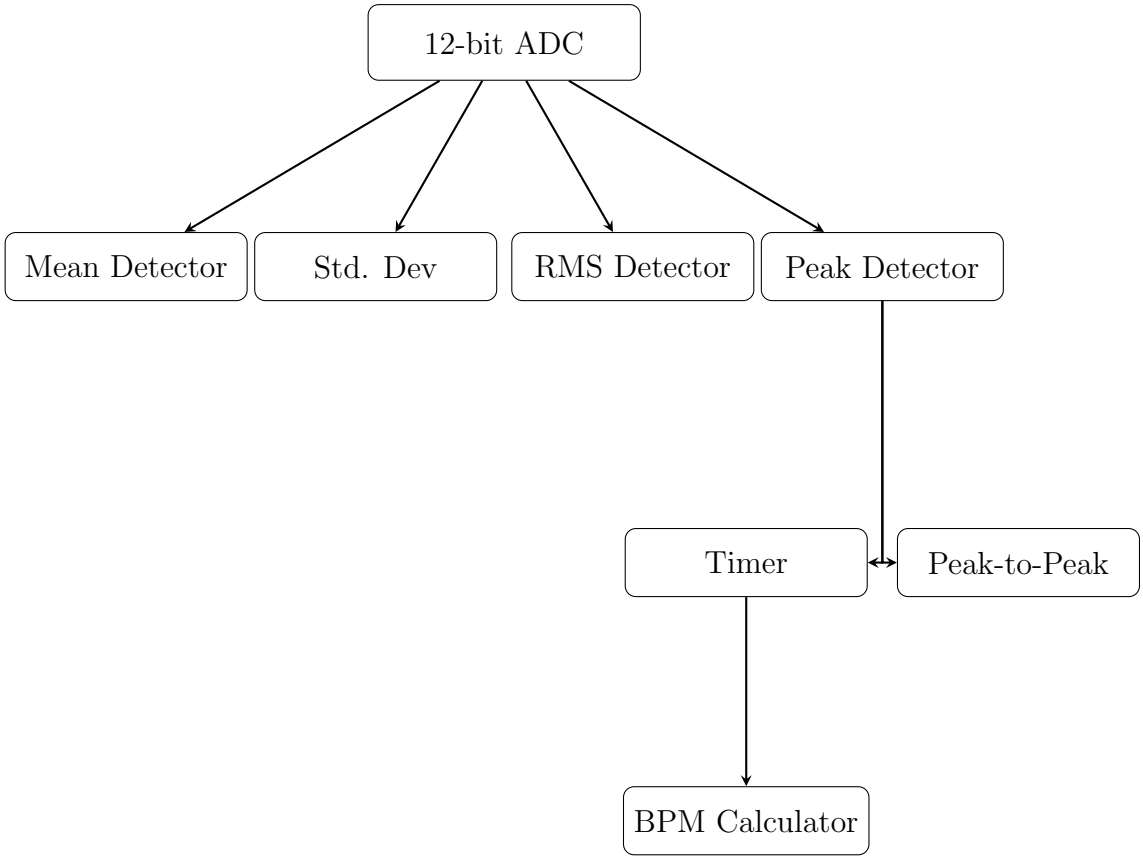

Figure 5.3: PPG Signal Processing Pipeline

5.4 Machine Learning Model Synthesis and Implementation

The Random Forest model architecture was systematically designed through iterative experimentation to identify the optimal configuration that maximizes diagnostic accuracy while maintaining feasibility for hardware implementation. The selected moderate configuration employs 30 decision trees with a maximum depth of 7 levels, minimum samples split of 15, and minimum samples leaf of 8. These hyperparameters were chosen based on cross-validation analysis that demonstrated diminishing returns in accuracy improvements beyond these values, while simultaneously considering the exponential growth in hardware resource requirements with increased tree count and depth. The feature selection process identified five critical PPG-derived parametersâmean amplitude, standard deviation, maximum amplitude, peak-to-peak variation, and root mean squareâeach contributing distinct information content for distinguishing between normal cardiac function and myocardial infarction patterns.

The model demonstrates exceptional classification performance across multiple evaluation metrics, as illustrated in Figure 5.4. The confusion matrix reveals strong discriminative capability with 252 true negatives and 245 true positives out of 516 test samples, yielding an overall accuracy of 96.32%. The model exhibits balanced performance between classes, with precision of 96.84% and recall of 95.70%, indicating minimal bias toward either the normal or pathological class. The F1-score of 96.27% further confirms the harmonic balance between

precision and recall. Notably, the Out-of-Bag (OOB) score of 96.46% provides an unbiased estimate of generalization performance, closely matching the test set accuracy and suggesting robust model training without significant overfitting. The feature importance analysis, shown in Figure 5.5, identifies standard deviation as the most discriminative parameter (importance score ≈ 0.30), followed by maximum amplitude (≈ 0.25) and mean amplitude (≈ 0.15), which aligns with physiological understanding that morphological variability in PPG waveforms is significantly altered during myocardial ischemic events.

Cross-validation analysis across five folds yielded scores ranging from 95.87% to 98.06% with a mean of 96.46% and standard deviation of 0.016, as depicted in the lower right panel of Figure 5.5. This narrow confidence interval (approximately $\pm 3.2\%$ at two standard deviations) demonstrates exceptional model stability across different data partitions, indicating that the classification performance is not dependent on specific training-test splits. The ROC analysis produced an AUC value of 0.9957, approaching the theoretical maximum and indicating near-perfect discrimination between the two classes across all decision thresholds. The model's macro and weighted averages both achieve 96% across precision, recall, and F1-score metrics, confirming that the balanced class distribution in the dataset (260 normal vs. 256 myocardial infarction samples) has been effectively leveraged without introducing class-specific biases. These comprehensive performance metrics validate the selected model configuration and feature set as optimal for the intended clinical diagnostic application, providing a robust foundation for real-time FPGA-based cardiac health monitoring systems.

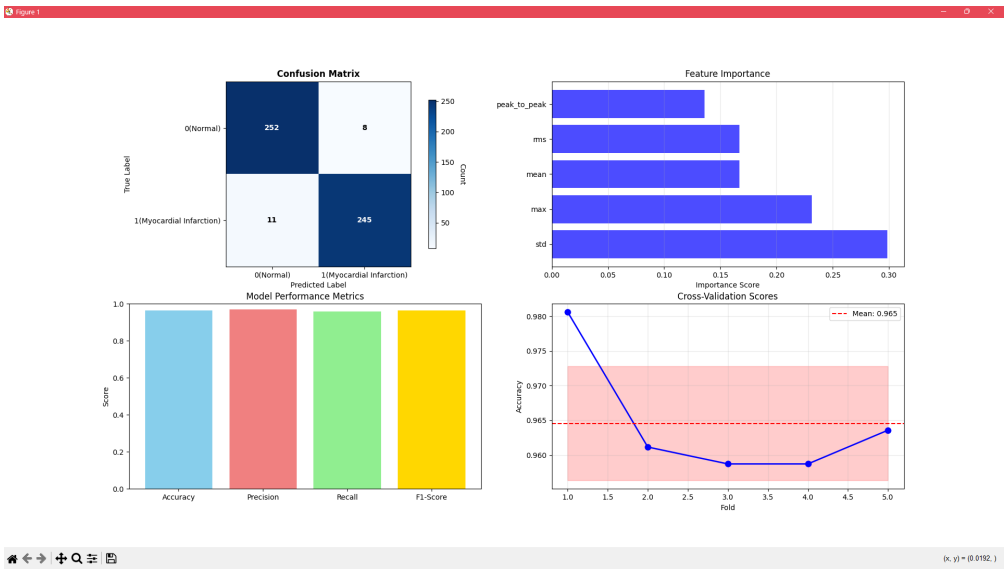

Figure 5.4: Confusion matrix and classification performance evaluation of the Random Forest model.

R

andom Forest ML IP Generation and Performance

The trained Random Forest machine learning model was converted into hardware-compatible C code using both HLS4ML and Conifer toolchains, enabling seamless integration into the Xilinx Vitis HLS workflow. Following C synthesis and RTL export, a hardware accelerator IP core was successfully generated, as shown in Figure 5.6. This custom IP leverages the parallel processing capabilities of FPGA logic to achieve extremely high throughput.

```
Train shape: (2060, 5)
Test shape: (516, 5)

======================================================================
SELECTED CONFIGURATION: MODERATE
======================================================================
Trees: 30
Max Depth: 7
Min Samples Split: 15
Min Samples Leaf: 8
======================================================================

*******************************************************

OOB SCORE: 0.96456

*******************************************************
Accuracy: 0.96318
Precision: 0.96838
Recall: 0.95703
F1 Score: 0.96267

Cross Validation Scores: [0.98058252 0.96116505 0.95873786 0.95873786 0.96359223]
Average CV Score: 0.96456 ± (0.01642)

Classification Report:

                          precision    recall  f1-score   support

               0(Normal)       0.96      0.97      0.96       260
1(Myocardial Infarction)       0.97      0.96      0.96       256

                accuracy                           0.96       516
               macro avg       0.96      0.96      0.96       516
            weighted avg       0.96      0.96      0.96       516
```

Figure 5.5: Feature importance analysis and cross-validation performance of the model.

The implemented classifier requires only 26 clock cycles to complete a full inference operation. At a system clock frequency of 100 MHz, this results in a total latency of approximately 260 ns per prediction. In comparison, the same model executed on a traditional Python-based inference engine typically requires multiple milliseconds to produce results. Therefore, the FPGA-based solution demonstrates a real-time performance improvement of over four orders of magnitude, making it highly suitable for continuous cardiac health monitoring applications.

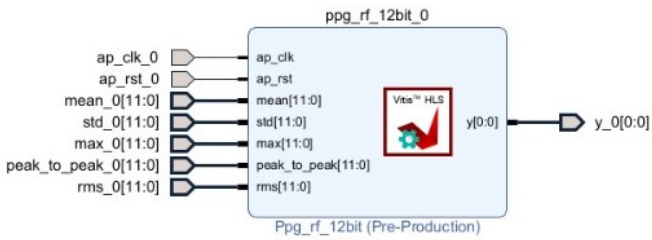

Figure 5.6: Random Forest classifier IP core for hardware acceleration.

5.5 Display and Validation

The extracted parameters are displayed on two LCD modules connected to the FPGA through their respective data and control pins (insert pin numbers). The first LCD presents the core cardiovascular indices: heart rate (in BPM), systolic amplitude, diastolic amplitude, and peak-to-peak value. The second LCD displays statistical descriptors of the waveform such as mean, RMS, and standard deviation, and additionally shows the classification output of the implemented ML model.

To further validate the system performance, multiple LEDs are incorporated to indicate key internal events such as peak detection and data-processing status. In parallel, raw PPG samples (with an appended safety bit) are transmitted over UART directly from the FPGA. These samples are acquired and decoded by a Python application running on a PC, which plots the live waveform for visual verification and debugging of the system.

5.6 Zero-PCB Interface Module

To provide a stable and clean interface between the PPG sensor, LCD modules, and the FPGA development board, a custom Zero-PCB interface module was designed and implemented. This board houses the TCRT1000 infrared PPG sensor and securely mounts the dual 16x2 LCD displays while routing their respective signal and power connections to the FPGA through well-organized headers. This approach significantly improves mechanical robustness, reduces loose wiring during testing, and ensures that signal integrity is maintained while handling the system.

The fabricated Zero-PCB module used for real-time validation experiments is shown in Figure 5.7.

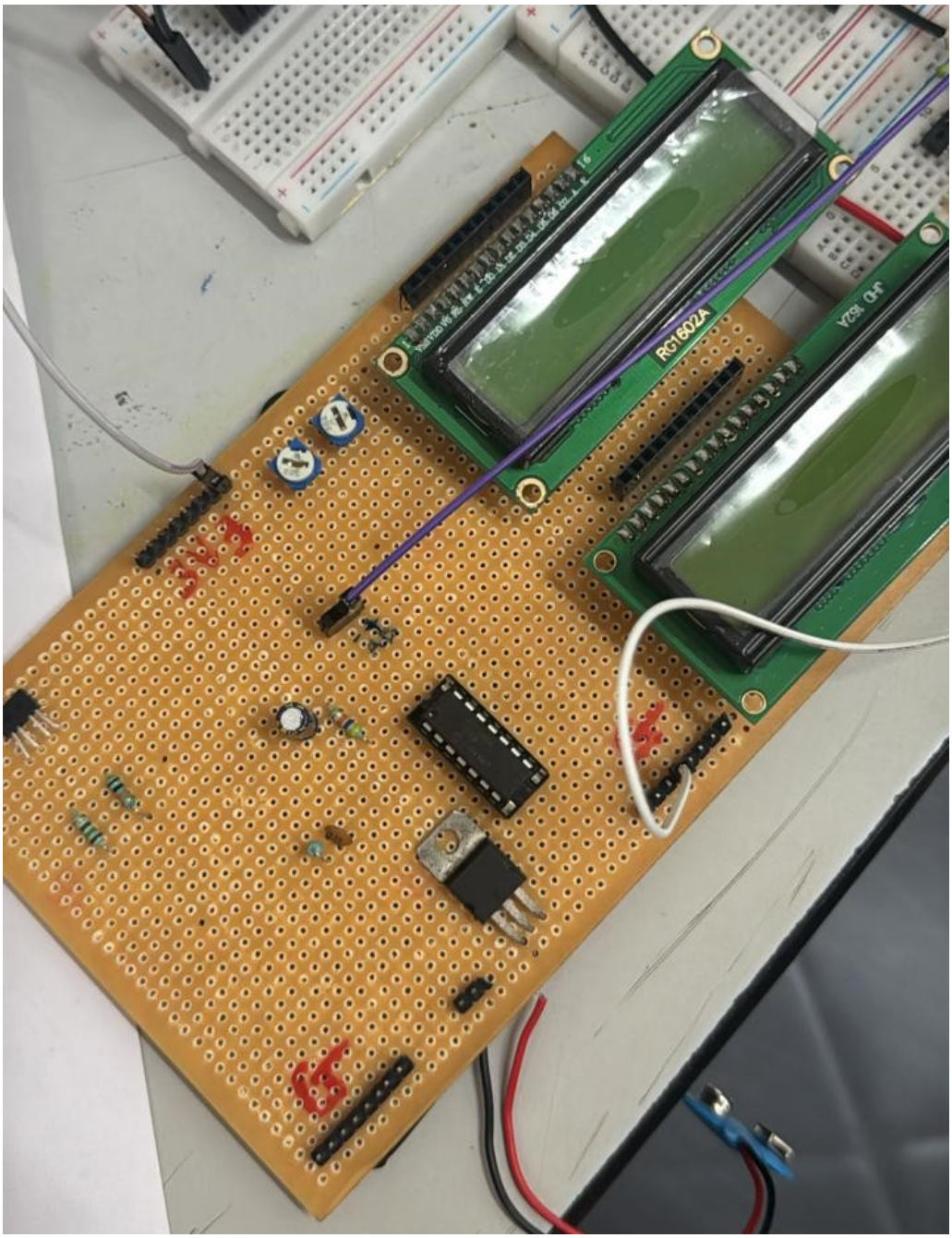

Figure 5.7: Zero-PCB interface developed to mount the PPG sensor and LCD displays while providing a clean connection to the FPGA board.

Chapter 6

Softwares

The development of this project required a combination of biomedical signal processing, digital hardware design, and machine learning implementation. The following software tools were used at various stages of the PPG waveform acquisition, conditioning, feature extraction, and disease prediction pipeline.

1. **Vivado / Vitis:** Vivado was used to design and synthesize the FPGA architecture that carries out real-time PPG signal processing. The custom modules for filtering, feature extraction, and UART transmission were written in Verilog and validated using Vivado simulation tools.

2. **Python:** Python served as the primary platform for validating the FPGA output, visualizing the live PPG waveform received via UART, and developing the neural network for disease prediction. It was also used to preprocess raw PPG data and extract statistical and morphological features required for training.

   - **NumPy:** Used for numerical operations such as vector arithmetic, filtering, and feature computation on large PPG datasets.
   - **Pandas:** Facilitated dataset structuring and data-cleaning pipelines required for machine learning training.
   - **SciPy:** Used for biomedical signal processing functions including denoising, peak detection, and interpolation.
   - **Scikit-Learn:** Enabled evaluation of classical ML models during experimentation (SVM, Random Forests) before migrating to neural networks.
   - **TensorFlow:** Used to develop and train the final neural network for heart disease prediction using extracted PPG features.
   - **HLS4ML:** Converted the trained neural network into a hardware-friendly representation suitable for execution on FPGAs.
   - **Conifer:** Generated optimized HDL logic for the neural network, allowing deployment inside the FPGA fabric with minimal latency.

3. **MATLAB:** MATLAB was used in the initial phase to model the PPG signal processing pipeline. Various digital filters were designed and tested on noisy datasets to evaluate their effects on the morphology of the PPG waveform. The *DSP System Toolbox* assisted in prototyping oversampling, decimation, peak detection, and detrending algorithms before porting them to FPGA hardware logic.

4. **LTSpice:** LTSpice was used to design and validate the analog front-end of the PPG sensor. Simulations were carried out for LED drivers, photodiode transimpedance amplifiers, and filtering networks to ensure adequate gain and noise performance before physically implementing the circuit.

Chapter 7

Simulation & Experimental Results

7.1 Analog PPG Signal Output

Before integrating the sensor into the digital acquisition pipeline, the TCRT-1000 module and its signal conditioning circuit were evaluated in analog mode to validate its ability to produce a usable PPG waveform. The output of the signal-conditioning stage (high-pass + low-pass filtering with amplification) was monitored on an oscilloscope while a team member placed a finger over the sensor.

The captured analog waveforms, shown in Figure 7.1, clearly demonstrates the characteristic morphology of a PPG signal, including the systolic peak, dicrotic notch, and diastolic component. The amplitude of the AC component falls well within the input range of the on-board ADC of the FPGA development board, confirming that the amplification and filtering is optimal before digitization. This verifies that the designed signal-conditioning circuit achieves its objective of removing the DC offset while preserving the pulsatile component corresponding to blood-volume changes.

Overall, the analog validation phase ensured that the sensorâanalog front-end subsystem produces a stable and noise-reduced PPG waveform suitable for reliable ADC sampling and digital processing in subsequent stages.

7.2 UART Streaming and Live Waveform Visualization

To verify that the digitized PPG samples were correctly acquired by the FPGA and transmitted without distortion, the sampled data was streamed over UART to a laptop. A custom Python script received the samples in real time, reconstructed them into a waveform, and plotted them on-screen. The plotted output (Figure 7.2) confirms that the raw PPG signal can be reliably sampled, transmitted and reconstructed, demonstrating preservation of waveform morphology. This validates the suitability of the UART data stream for subsequent operations such as feature extraction, peak detection and HRV analysis.

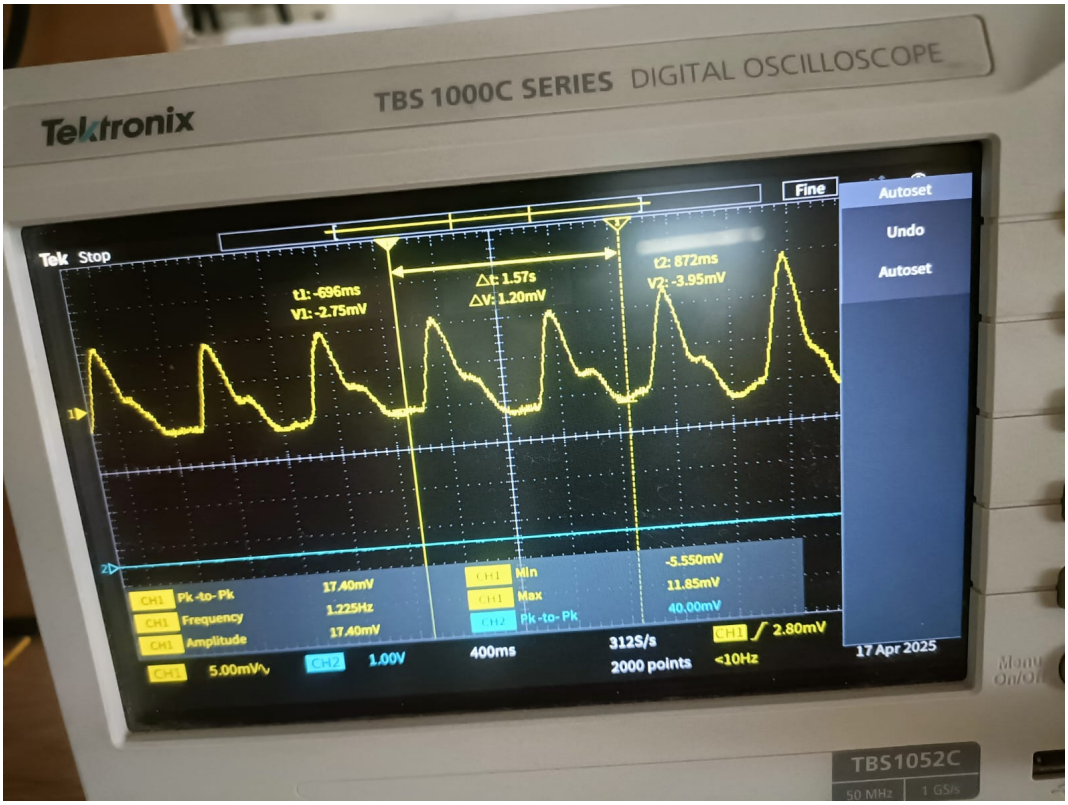

Figure 7.1: Captured PPG waveform

7.3 Results of Feature Extraction

The features extracted by the FPGA were displayed as 12-bit unsigned integers on the LCD screens. These parameters include heart rate (BPM), systolic amplitude, diastolic amplitude, peak-to-peak value, mean, RMS, and standard deviation. The complete processing flow was validated using both analog measurements and the UART data stream.

Figure 7.3 shows the analog PPG waveform captured directly on the Digital Storage Oscilloscope (DSO), confirming that the sensor and signal conditioning stages produce a clean, usable waveform.

Figure 7.4 displays the same PPG signal sampled by the 12-bit ADC and plotted over UART using Python. This validates that the ADC sampling rate, event-driven triggering, and UART transmission pipeline operate correctly and reliably.

Figures 7.5 and 7.6 show two instances of the FPGA performing real-time feature extraction. Each display includes accurate BPM estimates for the subject, along with systolic amplitude, diastolic amplitude, peak-to-peak value, mean, RMS, and standard deviation values. These confirm that the processing modules operate correctly and consistently across multiple test cycles.

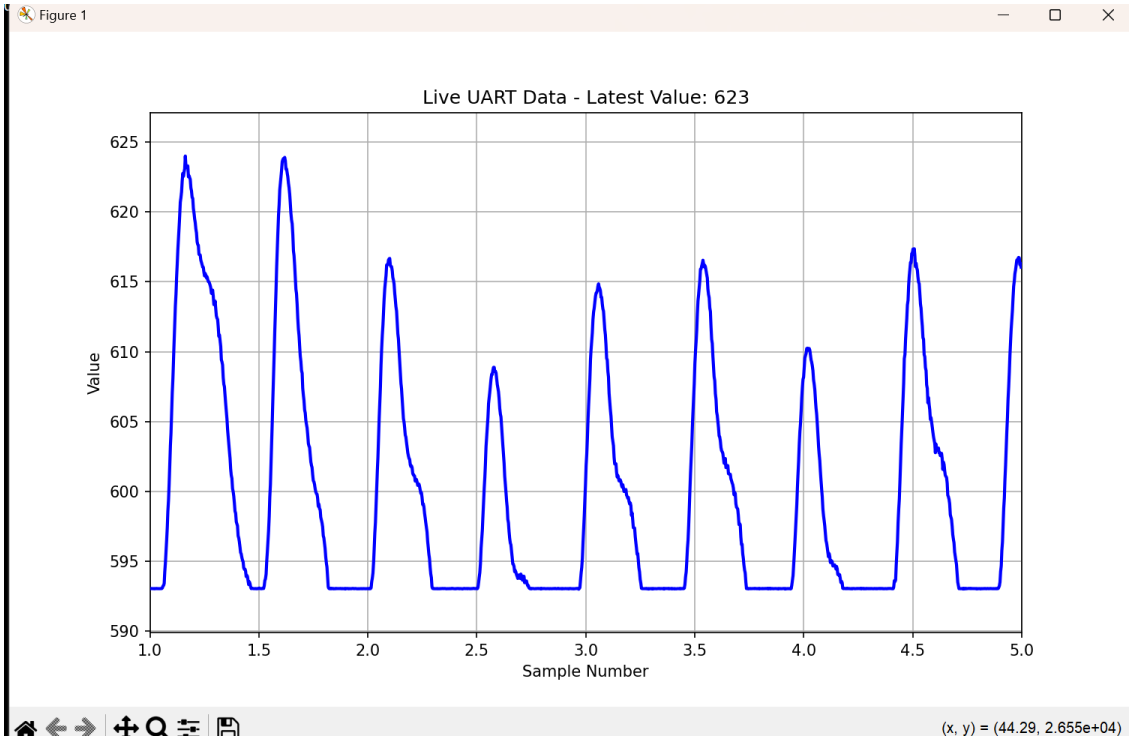

Figure 7.2: Live plotted raw PPG waveform reconstructed from UART stream on Python interface

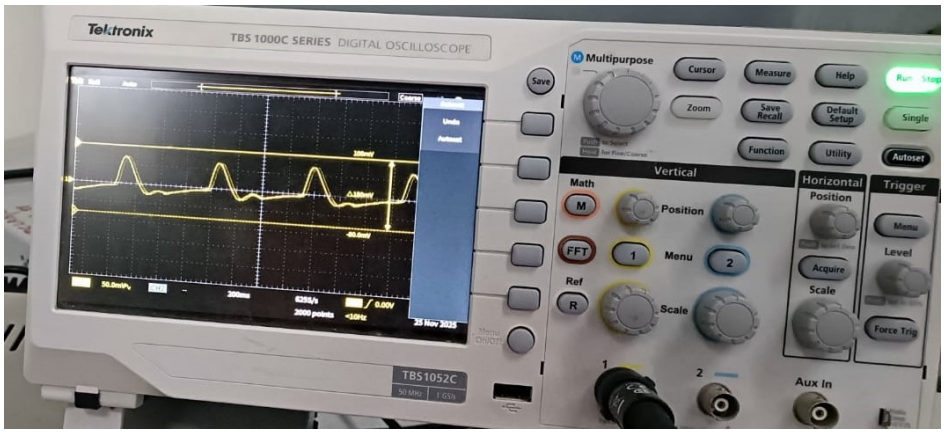

Figure 7.3: Analog PPG waveform captured from the DSO validating sensor and signal conditioning performance.

Figure 7.4: PPG waveform sampled by the ADC and transmitted over UART for validation.

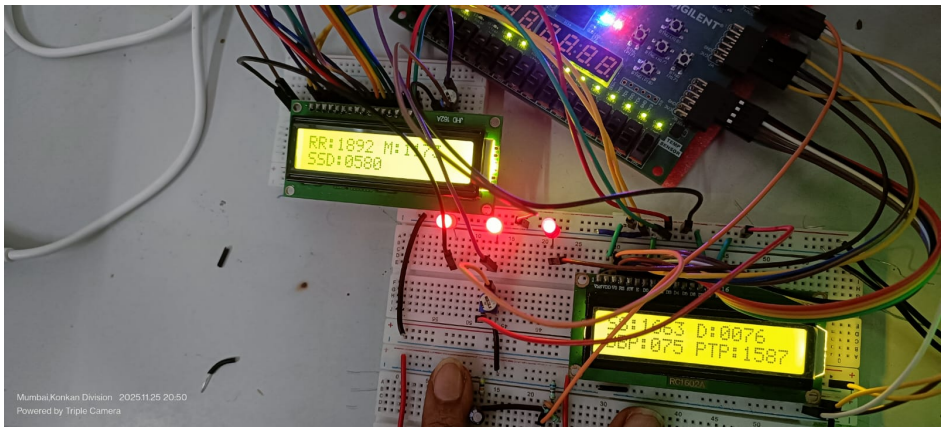

Figure 7.5: Feature extraction output showing BPM, systolic/diastolic peaks, and statistical parameters (Instance 1).

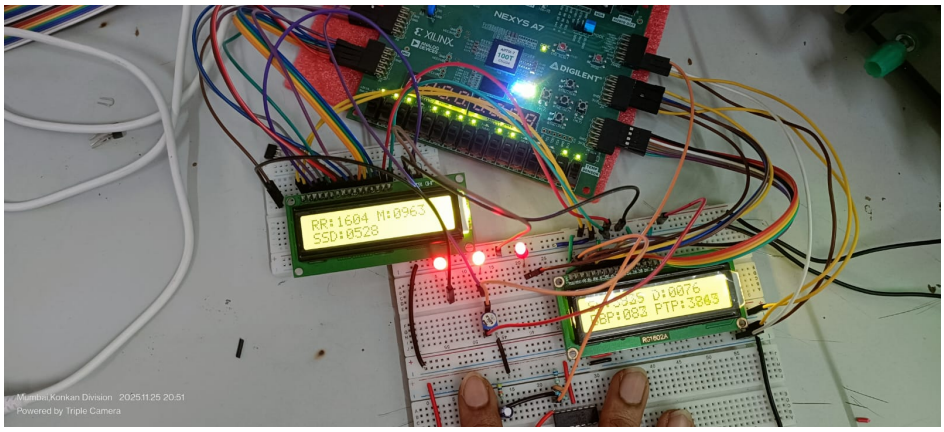

Figure 7.6: Feature extraction output showing BPM, systolic/diastolic peaks, and statistical parameters (Instance 2).

7.4 Integration with Machine Learning Model

7.4.1 ML IP Core Validation Through Simulation

The synthesized ML accelerator IP core was functionally verified using a comprehensive VHDL testbench within the Vivado simulation environment. Test vectors derived from the validation dataset were supplied to the IP as input feature samples, and the corresponding outputs were captured for waveform inspection. The timing results obtained from the simulation are shown in Figure 7.7.

As observed from the timing diagram, the model generates a valid classification result every 23 clock cycles. The binary decision output ("1" indicating myocardial infarction and "0" indicating a healthy state) is asserted once processing is completed. This validates correct functional behavior and confirms real-time inference capability of the deployed Random Forest classifier on FPGA hardware.

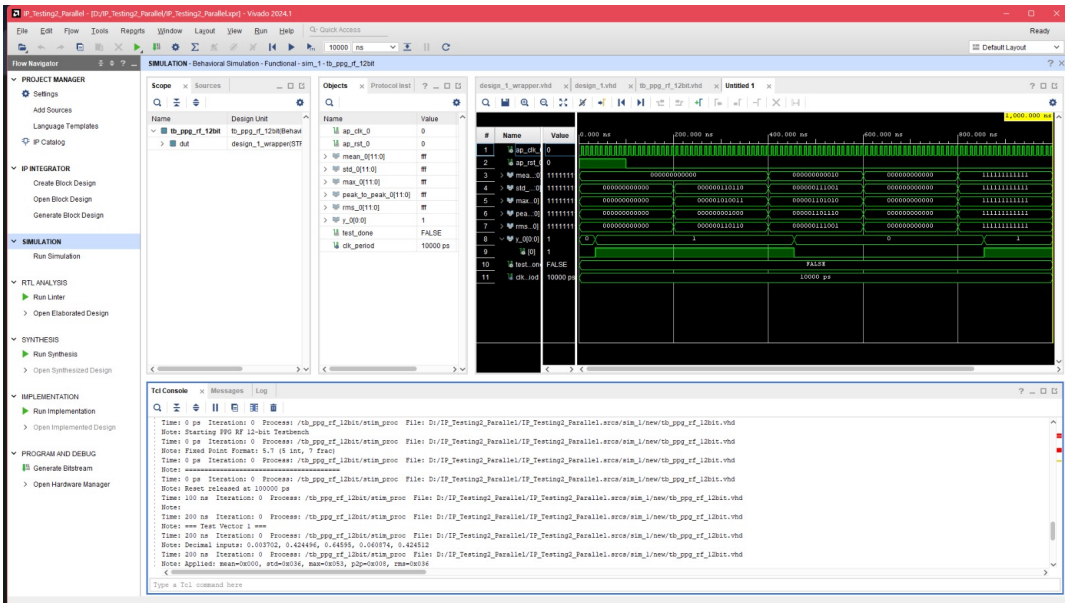

Figure 7.7: Timing diagram obtained through Vivado simulation illustrating the ML IP prediction output latency.

Chapter 8

Conclusion

This project demonstrates a fully functional and highly efficient FPGA-based system for real-time PPG waveform acquisition, cardiovascular feature extraction, and myocardial infarction prediction. By exploiting the inherent parallelism of FPGA architectures, the system overcomes the computational and latency constraints of conventional processor-based solutions used in portable biomedical devices.

The custom-designed TCRT1000 infrared PPG sensor and dedicated signal conditioning circuitry delivered high-fidelity waveforms with preserved pulsatile characteristics. Analog validation through oscilloscope testing and digital validation via UART streaming confirmed accurate signal acquisition and reliable ADC sampling without distortion.

All signal processing tasks were implemented in hardware using a modular design, enabling deterministic, real-time extraction of key cardiovascular features including heart rate, systolic and diastolic peak amplitudes, peak-to-peak variation, mean, RMS, and standard deviation. The adaptive multiscale peak detection algorithm ensured robust performance across variable physiological conditions.

A hardware-accelerated Random Forest classifier, optimized for 12-bit fixed-point inference, achieved an accuracy of 96.32%, AUC of 0.9957, and balanced precision-recall performance. With an execution latency of only 260 ns (26 clock cycles at 100 MHz), the model demonstrated more than four orders of magnitude speed improvement compared to software-based machine learning implementations, validating the suitability of FPGA acceleration for continuous cardiac monitoring.

Resource utilization analysis showed that only 34.16% of LUTs and 24.44% of flip-flops of the target Artix-7 device were consumed, leaving significant hardware margin for scalability. The customized zero-PCB interface ensured seamless integration of the sensor and displays, enhancing usability and system stability during testing.

Overall, this work successfully establishes FPGA-accelerated machine learning as a powerful and practical approach for next-generation, real-time biomedical diagnostics. The system is suitable for point-of-care deployment, wearable health devices, and remote patient monitoring applications where fast response, energy efficiency, and reliability are critical.

Future extensions of this work include:

- Integration of multimodal biosignals such as ECG and SpO_2 to enhance diagnostic accuracy
- Expansion to multi-class cardiac disease classification using deep learning accelerators
- Adaptive model retraining for personalized healthcare monitoring

- Comprehensive clinical validation with larger patient datasets

Through these advancements, the system can evolve into a compact, intelligent health monitoring platform capable of supporting early detection and preventive care for cardiovascular diseases.

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